SQL THROUGH ONE DATASET

DAY 11 / 15

06/08/26

I pivoted FY2020, FY2021 into columns on one row

Zero FY2020 sales nearly broke the growth math

ATLIQ HARDWARE   GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst

Problem Statement

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- Yesterday : Ranked products with RANK() OVER (
- PARTITION BY division)

- Leadership wants markets that grew FY2020 VS FY2021

The Trap ❌

- GROUP BY market , fiscal_year gives two seperate rows per market

Two rows per market , not one

can’t subtract FY2021- FY2020 across rows

Requirment

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FY2020 in its own column ✓

FY2021 in its own column ✓

Split the year first ,
then place side by side per market

The Fix

Use Window Function

```
SUM (CASE WHEN fiscal_year = 2020
THEN gross_sales ELSE 0 END ) as fy_2020_sales
```

A Quick Analogy

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Bank Statement

Every Transaction
months mixed in one long list
(GROUP BY month)

Split by month
Jan , Feb , March each get a column
(CASE WHEN pivot)

One row per account
every month now sits side by side
(single row output)

Compare two columns
This month vs last month = change
(Growth %)

Fix Query

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```
WITH gross_sales_report AS (
SELECT
 dc.customer_code , dc.customer ,
   dc.channel, dc.market, dd.fiscal_year,
 fsm.sold_quantity * fgp.gross_price AS gross_sales
   FROM fact_sales_monthly fsm
   JOIN dim_customer dc
 ON dc.customer_code = fsm.customer_code
 JOIN fact_gross_price fgp
 ON fgp.product_code = fsm.product_code
      AND fgp.fiscal_year = fsm.fiscal_year
 JOIN dim_date dd
 ON dd.calender_date = fsm.date
 WHERE dd.fiscal_year IN (2020,2021)),
```

Fix **Query**

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```
market_yearly_report AS (
SELECT gsr.market ,
SUM(CASE WHEN fiscal_year = 2020 THEN gross_sales ELSE 0 END)
AS fy_2020_sales,
SUM(CASE WHEN fiscal_year = 2021 THEN gross_sales ELSE 0 END)
AS fy_2021_sales
FROM gross_sales_report gsr
GROUP BY gsr.market
)

SELECT
myr.market,
ROUND(myr.fy_2020_sales/1000000,2) AS fy_2020_mln ,
ROUND(myr.fy_2021_sales/1000000,2) AS fy_2021_mln,
ROUND((myr.fy_2021_sales - myr.fy_2020_sales) /
NULLIF(myr.fy_2020_sales, 0) * 100,
2) AS grow_pct
FROM market_yearly_report myr
ORDER BY myr.market;
```

Results

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fact_sales_monthly | dim_customer | fact_gross_price | SQL File 5* | dim_date

Limit to 50000 rows

```sql
market_yearly_report AS (
SELECT gsr.market ,
SUM(CASE WHEN fiscal_year = 2020 THEN gross_sales ELSE 0 END) AS fy_2020_sales,
SUM(CASE WHEN fiscal_year = 2021 THEN gross_sales ELSE 0 END) AS fy_2021_sales
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

market	fy_2020_mln	fy_2021_mln	grow_pct
Australia	23.81	59.15	148.42
Austria	0.31	8.25	2589.72
Bangladesh	5.66	19.05	236.78
Brazil	2.33	2.14	-8.11
Canada	29.11	89.78	208.40
Chile	0.19	1.46	664.20
China	13.59	55.29	307.00
Columbia	0.03	0.37	1062.99
France	19.36	67.62	249.22
Germany	13.74	41.25	200.16
India	139.70	455.05	225.73
Indonesia	14.85	48.12	224.12
Italy	14.24	38.10	167.66
Japan	4.91	17.34	253.19

Result 11

Output

Action Output

#	Time	Action	Message
1	16:52:59	WITH gross_sales_report AS (SELECT dc.customer_code , dc.customer , dc.channel, dc.market, dd.fiscal_year, fs...	27 row(s) returned

CODE
BASICS

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| Vishnu Ram Sachin | Data Analyst

SQL THROUGH ONE DATASET CHALLENGE